

Student Name:
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Nonlinear Finite Element Method (Spring 2024)

Final Assignment

Due Date: 12:00pm, June 24, 2024

The final project accounts for 50% of your final grade. You may collaborate in groups of up to three students to develop the MATLAB code. However, you must conduct the analysis using commercial software independently and submit your own report without consulting other students.

1. **Problem 1** (50%) Implement the explicit Finite Element method in Matlab for stress analysis of 2D dynamic finite deformation problems by completing the predefined Matlab functions. Suppose the material follows the J2 isotropic plasticity law with linear hardening, the yield stress is given by,

$$\sigma_c = \sigma_0(1 + A\bar{\epsilon}_p)^\alpha$$

where σ_c , σ_0 , A , $\bar{\epsilon}_p$ and α are the yield stress, the initial yield stress, a constant, effective von-mises plastic strain, and power, respectively.

Using the Hencky strain in the constitutive law instead of the deformation gradient, i.e.,

$$\epsilon_{ij} = \frac{1}{2} \log C_{ij}$$

where C_{ij} is the right cauchy green deformation tensor, the total strain of the material may be decomposed additively into the elastic and plastic part,

$$\epsilon_{ij} = \epsilon_{ij}^e + \epsilon_{ij}^p$$

and the von-mises effective plastic strain is defined as

$$\bar{\epsilon}_p = \sqrt{\frac{2}{3} \epsilon_{ij}^p \epsilon_{ij}^p}$$

Finally, the stress may be calculated by

$$\sigma_{ij} = K \epsilon_{kk} \delta_{ij} + 2G \epsilon_{ij}^{e, \text{dev}}$$

where K and G are the bulk modulus and shear modulus of the material, respectively. The deviatoric part of the elastic strain tensor is given by

$$\epsilon_{ij}^{e, \text{dev}} = \epsilon_{ij}^e - \frac{1}{3} \epsilon_{kk} \delta_{ij}$$

- (a) Functions in *main_dynamics.m*, *generate_mesh.m*, *g_center.m*, *plot_dyn_results.m*, *quadplot.m*, and *shape_function.m* are completed. *main_dynamics.m* is the main program of the Matlab code, which shows the steps for a Finite Element analysis process. Read the code starting from *main_dynamics.m* to understand the meaning of data structures and follow the calling order of the subroutines.

- (b) Implement the functions marked with "TODO" comments following the instructions in the code:

lumped_mass.m : formation of the lumped mass matrix
consistent_mass.m : formation of the consistent mass matrix
boundary_coditions.m : set up the displacement boundary conditions and initial conditions
time_step.m : determine the critical time step of the analysis
explicit_solver.m : central difference time integrator
constitutive.m : compute the stress according to J2 isotropic plasticity law
fint_vector.m : formation of the internal nodal force vector

Note that the important data structures defined in the Matlab code are

$$x_a = \begin{pmatrix} x^{(1)} & y^{(1)} \\ x^{(2)} & y^{(2)} \\ \vdots & \vdots \\ x^{(N)} & y^{(N)} \end{pmatrix}, \quad elem = \begin{pmatrix} I^{(1)} & J^{(1)} & \dots & K^{(1)} \\ I^{(2)} & J^{(2)} & \dots & K^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ I^{(M)} & J^{(M)} & \dots & K^{(M)} \end{pmatrix}, \quad xg = \begin{pmatrix} x^{(1)} & y^{(1)} \\ x^{(2)} & y^{(2)} \\ \vdots & \vdots \\ x^{(M)} & y^{(M)} \end{pmatrix}.$$

where $x^{(i)}$ and $y^{(i)}$ in x_a are the coordinates of node $i \in \{1, 2, \dots, N\}$, $\{I^{(q)}, J^{(q)}, \dots, K^{(q)}\}$ are the indices of the nodes of element $q \in \{1, 2, \dots, M\}$, i.e. $elem$ is the connectivity table, and $x^{(q)}$ and $y^{(q)}$ in xg are the coordinates of the barycenter of element q .

2. **Problem 2** (50%) Consider a dynamic elastoplastic problem on the trapezoidal panel domain as shown in Figure 2. The vertical left edge is fixed and all other edges are traction free, i.e. $\bar{\mathbf{t}} = \mathbf{0}$.

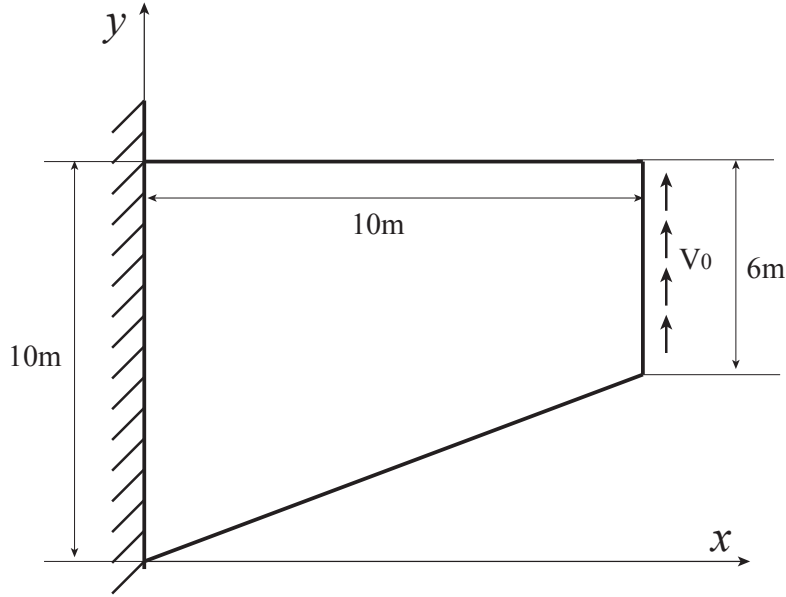


Figure 1: Cooke's membrane configuration.

An initial velocity v_0 in the vertical direction is applied to the right edge of the domain. Material properties for the panel are listed in Table 1.

- (a) Use your matlab code developed in Problem 1 to solve the problem with initial conditions $v_0 = 10$ and 100 m/s, respectively. Plot the deformed panel showing the von-mises stress distribution of the domain in your report. Plot the stress-strain curve at an element which experienced plastic deformation.

Young's modulus (GPa)	Density (Kg/m ³)	Initial yield stress (MPa)	Constant A	Power
100	8950	400	10	0.25

Table 1: Material properties

(b) Solve the problem in Ansys or other commercial software and validate your solution from (a) against the one obtained in commercial software by comparing the von-mises stress distribution, effective plastic strain and displacement field.